Tax Pass-Through and Incidence: Monopoly (Weyl and Fabinger, 2013)

Draft notes

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Abstract

Self-contained notes on pass-through and tax incidence, following Weyl and Fabinger (2013) with full derivations under perfect competition and monopoly (Sections 2–3). The extension to product returns is in `incidence_under_return.tex`.

1 Shared notation

- $Q = D(p)$: demand at consumer price p, with $D' < 0$.
- $\varepsilon_D \equiv -pD'(p)/D(p) > 0$: price elasticity of demand.
- t or τ: per-unit tax or cost shock borne by the seller.
- $\rho \equiv dp/d\tau$: **pass-through** (consumer price rise per dollar of seller cost).
- $CS(p) = \int_p^{\bar{p}} D(x)\,dx$, π: consumer surplus and profit.
- $I \equiv |\Delta CS|/|\Delta\pi|$: **incidence ratio** (monopoly); under perfect competition $I = \rho/(1-\rho)$.

2 Perfect competition: Weyl and Fabinger, Section II

Consumers pay p; producers receive $p - \tau$ when a per-unit tax τ is levied on producers. Market clearing is

$$D(p) = S(p - \tau). \tag{1}$$

Let Q denote equilibrium quantity and $\rho \equiv dp/d\tau$.

2.1 Principle 1: Physical incidence is neutral

Same equilibrium, different labels. Whether the per-unit levy is legally collected from producers or from consumers, take p to be the price consumers pay and $p - \tau$ the amount producers receive. Then (1) is unchanged and the equilibrium (p, Q) is the same; only the wording of who remits τ differs.

When τ rises by $\Delta\tau$. Along the equilibrium path, the consumer price always rises by $\rho\,\Delta\tau$ (definition of pass-through). Producer net receipt $u \equiv p - \tau$ changes by

$$\Delta u = \Delta p - \Delta\tau = (\rho - 1)\,\Delta\tau = -(1-\rho)\,\Delta\tau.$$

So the producer bears $(1-\rho)$ of each dollar of tax at the margin; the consumer bears ρ through the higher price.

Two ways to describe the same marginal change. It is easy to mix up *which* price moves when the tax is labeled differently:

- **Producer tax** (this section's default): τ is subtracted from what producers keep. A rise $\Delta\tau$ pushes the *consumer* price up by $\rho\,\Delta\tau$.
- **Consumer tax** (equivalent accounting): split the consumer payment into a *list price* $\tilde{p}$ plus the tax, so consumers pay $\tilde{p}+\tau$ and producers receive $\tilde{p}$. At the same equilibrium, $\tilde{p} = p - \tau$. When τ rises by $\Delta\tau$, the total consumer payment still rises by $\rho\,\Delta\tau$, so the list price must *fall* by $(1-\rho)\,\Delta\tau$—not by the full $\Delta\tau$. Producer receipt ($\tilde{p}$) falls by that same $(1-\rho)\,\Delta\tau$.

In either description, producer net receipt falls by $(1-\rho)\,\Delta\tau$ while the consumer payment rises by $\rho\,\Delta\tau$.

2.2 Principles 2–3: Burden sharing and incidence

Consumer surplus $CS(p) = \int_p^{\bar{p}} D(x)\,dx$ satisfies $dCS = -D(p)\,dp$, so

$$\frac{dCS}{d\tau} = -D(p)\,\rho. \tag{2}$$

Producer surplus is $PS(u)$ with $u \equiv p - \tau$ and $PS(u) = \int_0^u S(x)\,dx$, so $PS'(u) = S(u) = Q$. Since $u = p - \tau$,

$$\frac{dPS}{d\tau} = PS'(u)\left(\frac{dp}{d\tau} - 1\right) = Q(\rho - 1) = -(1-\rho)\,Q.$$

Principle 2: the mechanical tax bill $Q\,d\tau$ splits with weights ρ on consumers and $1-\rho$ on producers.

Principle 3:

$$I \equiv \frac{|dCS/d\tau|}{|dPS/d\tau|} = \frac{\rho Q}{(1-\rho)Q} = \frac{\rho}{1-\rho}.$$

2.3 Principle 4: Pass-through formula

Differentiate (1) with respect to τ:

$$D'(p)\,\frac{dp}{d\tau} = S'(p-\tau)\left(\frac{dp}{d\tau} - 1\right).$$

Collect terms:

$$\left[S'(p-\tau) - D'(p)\right]\rho = S'(p-\tau) \quad \Rightarrow \quad \rho = \frac{S'(p-\tau)}{S'(p-\tau) - D'(p)}.$$

Define elasticities at the equilibrium (Weyl and Fabinger 2013, p. 534):

$$\varepsilon_D \equiv -\frac{p\,D'(p)}{Q} > 0, \qquad \varepsilon_S \equiv \frac{p\,S'(p-\tau)}{Q} > 0.$$

Since $Q = D(p) = S(p - \tau)$, multiply numerator and denominator of $\rho = S'/(S' - D')$ by p/Q:

$$\rho = \frac{\varepsilon_S}{\varepsilon_S + \varepsilon_D} = \frac{1}{1 + \varepsilon_D/\varepsilon_S}.$$

The inelastic side bears more of the tax.

2.4 Principle 5: Local to global

For a finite tax change $\tau_0 \to \tau_1$, integrate (2)–(2.2):

$$\Delta CS = -\int_{\tau_0}^{\tau_1} \rho(\tau)\, Q(\tau)\, d\tau, \qquad \Delta PS = -\int_{\tau_0}^{\tau_1} (1 - \rho(\tau))\, Q(\tau)\, d\tau.$$

Define the quantity-weighted average pass-through

$$\bar{\rho}_{\tau_0}^{\tau_1} \equiv \frac{\int_{\tau_0}^{\tau_1} \rho(\tau)\, Q(\tau)\, d\tau}{\int_{\tau_0}^{\tau_1} Q(\tau)\, d\tau}.$$

Then

$$I_{\tau_0}^{\tau_1} = \frac{\Delta CS}{\Delta PS} = \frac{\bar{\rho}_{\tau_0}^{\tau_1}}{1 - \bar{\rho}_{\tau_0}^{\tau_1}}.$$

The local formula $I = \rho/(1 - \rho)$ holds globally once ρ is replaced by $\bar{\rho}$. If τ rises until $Q = 0$, the ratio CS/PS of total surpluses equals $\bar{\rho}/(1 - \bar{\rho})$ with $\bar{\rho}$ averaged from $\tau = 0$ to the choking tax.

3 Monopoly: Weyl and Fabinger, Section III

The monopolist chooses quantity Q facing inverse demand $p(Q)$ with $p'(Q) < 0$. Revenue is $p(Q)\, Q$, cost is $c(Q)$ with $\mathrm{MC}(Q) \equiv c'(Q)$. Define

$$\mathrm{MR}(Q) \equiv \frac{d}{dQ}\big[p(Q)\, Q\big] = p(Q) + p'(Q)\, Q,$$

$$m(Q) \equiv p(Q) - \mathrm{MC}(Q), \qquad ms(Q) \equiv -p'(Q)\, Q \quad \text{(marginal consumer surplus)}.$$

With a per-unit producer tax τ, the FOC is

$$\mathrm{MR}(Q) = \mathrm{MC}(Q) + \tau. \tag{3}$$

We write p, Q, m, ms evaluated at the optimum unless noted.

3.1 Principle 1: Physical incidence is neutral

Producer tax. Problem: $\max_Q \big[p(Q) - \mathrm{MC}(Q) - \tau\big]\, Q$. FOC: $\mathrm{MR}(Q) = \mathrm{MC}(Q) + \tau$.

Consumer tax. Consumers pay p; the monopolist keeps $p - \tau$. Problem: $\max_Q \big[p(Q) - \tau - \mathrm{MC}(Q)\big]\, Q$. The FOC is identical.

Conclusion. Only $p - \tau - \mathrm{MC}$ matters for Q. Physical incidence does not affect real allocation; $\rho = dp/d\tau$ is the same under either labeling.

3.2 Principle 2: Total burden exceeds tax revenue

Step 1: Consumer burden. Demand is still $Q = D(p)$, so as in (2),

$$\frac{dCS}{d\tau} = -Q\,\rho. \tag{4}$$

Step 2: Producer burden (envelope theorem). Profit is $\pi(Q,\tau) = \big[p(Q) - \mathrm{MC}(Q) - \tau\big]\,Q$. Holding Q fixed,

$$\frac{\partial \pi}{\partial \tau} = -Q.$$

At the optimum, $\partial\pi/\partial Q = 0$, so the total derivative equals the partial derivative:

$$\frac{d\pi}{d\tau} = -Q. \tag{5}$$

The monopolist bears the full mechanical tax $Q\,d\tau$ at the margin.

Step 3: Compare to tax revenue. Revenue raised is $Q\,d\tau$. Total burden on participants:

$$|\Delta CS| + |\Delta\pi| = \rho Q\,d\tau + Q\,d\tau = (1+\rho)\,Q\,d\tau > Q\,d\tau.$$

Under perfect competition (2.2), producers bore $(1-\rho)Q\,d\tau$ and the total equaled revenue. Under monopoly the extra $\rho Q\,d\tau$ is **excess burden** on consumers from the quantity distortion.

3.3 Principle 3: Incidence equals pass-through

From (4)–(5),

$$I \equiv \frac{|\Delta CS|}{|\Delta\pi|} = \frac{\rho Q}{Q} = \rho.$$

The monopolist pays the tax dollar-for-dollar (at fixed Q); any consumer loss beyond mechanical shifting is excess burden, and its ratio to producer loss is ρ.

3.4 Social incidence of competition (constant marginal cost)

Assume $\mathrm{MC} \equiv c$. A quantity $\tilde{q}$ is supplied competitively at cost c. Incumbent profit: $\pi = \big[p(Q) - c\big](Q - \tilde{q})$.

Step 1: Competition mimics a cost shock. Marginal revenue becomes $p(Q) + p'(Q)(Q - \tilde{q})$. Increasing $\tilde{q}$ by $d\tilde{q}$ shifts the FOC exactly like raising τ by $-p'(Q)\,d\tilde{q}$:

$$\frac{dQ}{d\tilde{q}} = p'(Q)\,\frac{dQ}{d\tau} = \frac{p'(Q)}{\mathrm{MR}'} = \rho.$$

Step 2: Deadweight loss. $m(Q) = p(Q) - c$. Social optimum Q^{**} satisfies $m(Q^{**}) = 0$.

$$\mathrm{DWL}(Q) = \int_Q^{Q^{**}} m(q)\,dq \quad \Rightarrow \quad \frac{d\,\mathrm{DWL}}{dQ} = -m(Q).$$

Hence

$$\frac{d\,\mathrm{DWL}}{d\tilde{q}} = \frac{d\,\mathrm{DWL}}{dQ}\,\frac{dQ}{d\tilde{q}} = -m\,\rho.$$

Step 3: Profit loss. Envelope theorem at fixed Q: $d\pi/d\tilde{q} = -m$.

Step 4: Social incidence.

$$\mathrm{SI} \equiv \frac{d\,\mathrm{DWL}/d\tilde{q}}{d\pi/d\tilde{q}} = \frac{-\rho m}{-m} = \rho.$$

Competition erodes deadweight loss and profit at the same rate as consumer incidence of a tax.

3.5 Principle 4: Pass-through formula

Step 1: Differentiate the FOC. From (3), $\mathrm{MR}(Q) = \mathrm{MC}(Q) + \tau$. Differentiate with respect to τ:

$$\mathrm{MR}'(Q)\,\frac{dQ}{d\tau} = \mathrm{MC}'(Q)\,\frac{dQ}{d\tau} + 1 \quad \Rightarrow \quad \left[\mathrm{MR}' - \mathrm{MC}'\right]\frac{dQ}{d\tau} = 1.$$

Hence

$$\frac{dQ}{d\tau} = \frac{1}{\mathrm{MR}' - \mathrm{MC}'}.$$

Step 2: Pass-through in price. Consumer price is $p(Q)$, so

$$\rho = \frac{dp}{d\tau} = p'(Q)\,\frac{dQ}{d\tau} = \frac{p'(Q)}{\mathrm{MR}' - \mathrm{MC}'}. \tag{6}$$

Step 3: Constant marginal cost ($\mathrm{MC}' = 0$). $\mathrm{MR} = p + p'Q$, so

$$\mathrm{MR}' = p' + p' + p''Q = 2p' + p''Q,$$

and (6) becomes

$$\rho = \frac{p'}{2p' + p''Q} = \frac{1}{2 + p''Q/p'}. \tag{7}$$

Step 4: Introduce ε_{ms}. Since $ms = -p'Q$,

$$ms' = \frac{d}{dQ}(-p'Q) = -p' - p''Q \quad \Rightarrow \quad \frac{p''Q}{p'} = -\frac{ms'}{p'} - 1.$$

Define

$$\varepsilon_{ms} \equiv \frac{ms}{ms'\,Q} \quad \Rightarrow \quad \frac{1}{\varepsilon_{ms}} = \frac{ms'\,Q}{ms} = \frac{p''Q}{p'} + 1,$$

so $p''Q/p' = 1/\varepsilon_{ms} - 1$. Substitute into (7):

$$\rho = \frac{1}{2 + 1/\varepsilon_{ms} - 1} = \frac{1}{1 + 1/\varepsilon_{ms}}.$$

This is the *curvature* part; supply and demand elasticities enter next.

Step 5: Introduce ε_D and ε_S. Write (6) as $\rho = 1/\big(\mathrm{MR}'/p' - \mathrm{MC}'/p'\big)$. Since $\mathrm{MR} = p - ms$ with $ms \equiv -p'Q$,

$$\mathrm{MR}' = p' - ms' \quad \Rightarrow \quad \frac{\mathrm{MR}'}{p'} = 1 - \frac{ms'}{p'} = 2 + \frac{p''Q}{p'} = 1 + \frac{1}{\varepsilon_{ms}},$$

using $ms'/p' = -1 - p''Q/p'$ and $p''Q/p' = 1/\varepsilon_{ms} - 1$. For the cost side, with $\varepsilon_S \equiv \mathrm{MC}/(\mathrm{MC}'\,Q)$,

$$\frac{\mathrm{MC}'}{p'} = \frac{1}{\varepsilon_S\,Q} \cdot \frac{\mathrm{MC}}{p'}.$$

At $\tau = 0$, Lerner's rule $\mathrm{MC}/p = (\varepsilon_D - 1)/\varepsilon_D$ and $\varepsilon_D = -p/(Qp')$ imply

$$\frac{\mathrm{MC}}{p'} = \frac{\mathrm{MC}}{p} \cdot \frac{p}{p'} = -\frac{(\varepsilon_D - 1)\,Q}{p}, \qquad \frac{\mathrm{MC}'}{p'} = -\frac{\varepsilon_D - 1}{\varepsilon_S}.$$

Therefore

$$\frac{\mathrm{MR}'}{p'} - \frac{\mathrm{MC}'}{p'} = 1 + \frac{1}{\varepsilon_{ms}} + \frac{\varepsilon_D - 1}{\varepsilon_S},$$

and

$$\rho = \frac{1}{1 + \dfrac{\varepsilon_D - 1}{\varepsilon_S} + \dfrac{1}{\varepsilon_{ms}}}. \tag{8}$$

When $\mathrm{MC}' = 0$ (constant marginal cost), $\varepsilon_S \to \infty$ and (8) reduces to $\rho = 1/(1 + 1/\varepsilon_{ms})$ from Step 4.

Step 6: Interpretation.

- **vs. perfect competition:** ε_D becomes $\varepsilon_D - 1$ because monopoly requires $\varepsilon_D > 1$.
- **Curvature:** $1/\varepsilon_{ms}$ comes from p''. Log-concave demand ($\phi'' < 0$ for $\phi = \log D$) implies $\rho > 1$ when MC is constant.
- **Linear inverse demand** $p(Q) = a - bQ$: $p'' = 0$, $\mathrm{MR}' = -2b$, $p' = -b$, so $\rho = 1/2$.

3.6 Principle 5: Local to global

Throughout this section, τ denotes the **per-unit tax level** (a cost shifter in (3)), not the consumer price p. Pass-through $\rho(\tau) \equiv dp/d\tau$ and quantity $Q(\tau)$ are evaluated along the equilibrium path as τ varies.

Integrating over a tax change. For a finite tax change $\tau_0 \to \tau_1$, integrate (4)–(5) as in Section 2:

$$\Delta CS = -\int_{\tau_0}^{\tau_1} \rho(\tau)\,Q(\tau)\,d\tau, \qquad \Delta\pi = -\int_{\tau_0}^{\tau_1} Q(\tau)\,d\tau.$$

With $\bar{\rho}$ the quantity-weighted average pass-through,

$$\bar{\rho}_{\tau_0}^{\tau_1} \equiv \frac{\int_{\tau_0}^{\tau_1} \rho(\tau)\,Q(\tau)\,d\tau}{\int_{\tau_0}^{\tau_1} Q(\tau)\,d\tau},$$

the global incidence ratio is $I = \bar{\rho}$ (not $\bar{\rho}/(1-\bar{\rho})$ as under perfect competition).

Social incidence: a different shock index. The competition exercise in the previous subsection is *not* indexed by τ. Competitive supply $\tilde{q} \in [0, Q^{**}]$ plays the role of the shock; pass-through $\rho(\tilde{q})$ is the consumer-price response to marginal entry. Average with weight $m(\tilde{q})$:

$$\tilde{\rho} \equiv \frac{\int_0^{Q^{**}} \rho(\tilde{q})\, m(\tilde{q})\, d\tilde{q}}{\int_0^{Q^{**}} m(\tilde{q})\, d\tilde{q}}.$$

Then $\mathrm{DWL}/\pi = \tilde{\rho}$ globally (both numerator and denominator vanish at $\tilde{q} = Q^{**}$).

3.7 Summary: perfect competition vs. monopoly

	Perfect competition	Monopoly
$dCS/d\tau$	$-\rho Q$	$-\rho Q$
$d\pi/d\tau$	$-(1-\rho)Q$	$-Q$
I	$\rho/(1-\rho)$	ρ
ρ	$1/(1+\varepsilon_D/\varepsilon_S)$	$1/(1+\frac{\varepsilon_D-1}{\varepsilon_S}+\frac{1}{\varepsilon_{ms}})$

References

References

[Weyl and Fabinger(2013)] Weyl, E. G. and M. Fabinger (2013). Pass-through as an economic tool: Principles of incidence under imperfect competition. *Journal of Political Economy*, 121(3):528–583.